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1. Literature Review

1.1 Introduction

E-commerce has become a central component of global retail. Brazil is Latin America's largest e-commerce market, accounting for roughly 42% of the region's B2C activity, and has about 150 million internet users (da Veiga et al., 2024). Its continued growth raises important questions about both order fulfilment and the customer's post-purchase experience. Deshpande and Pendem (2023) find that customers assess delivery relative to the promised date, penalising lateness more than they reward early arrival, and that these logistics ratings subsequently affect purchasing and seller sales. Cui et al. (2024) add a related finding: changing the promised delivery speed alone, independent of actual delivery performance, can shift sales, returns and satisfaction, indicating that customers respond to delivery expectations as well as delivery outcomes. These findings identify fulfilment dynamics, customer expectations and post-order satisfaction as promising areas for exploration. This review examines relevant modelling approaches, informative features and evaluation methods to guide the subsequent exploratory analysis and the problem identification.

1.2 Existing Literature Works

Research on e-commerce post-order outcomes covers both operational forecasting and customer evaluation. Salari et al. (2022) use JD.com data to forecast delivery times and set delivery promises that account for the unequal costs of overpromising and underpromising, estimating a 6.1% increase in sales. Kandula et al. (2021) predict last-mile delivery success to optimize schedules, yielding estimated cost savings of 7.2%-10.2% across 2 hubs. Together, these studies show how predictive models can support different operational decisions.

Delivery performance also affects how customers rate their experience. Using 93,131 orders from a South American e-commerce marketplace, Akturk et al. (2022) find that longer processing and delivery times lower ratings, with processing time having the stronger effect, and that lateness is penalised more heavily for high-value orders. Both early and late delivery show nonlinear effects, indicating that ratings reflect fulfilment performance and transaction context rather than delivery status alone.

1.3 Modeling Approaches

Large-scale benchmark studies place tree-based ensembles among the strongest methods for structured data. Across 45 classification and regression datasets, Grinsztajn et al. (2022) found that tree-based models remained state-of-the-art on medium-sized datasets even after extensive hyperparameter tuning, and that they were particularly robust to uninformative variables and irregular threshold-based relationships in individual features. McElfresh et al. (2023) extended this comparison to 19 algorithms across 176 datasets and found CatBoost and XGBoost achieving the strongest average ranks, ahead of Random Forest and further ahead of logistic regression. However, nearly every algorithm ranked first on at least one dataset and last on another, and for roughly one-third of the datasets, light hyperparameter tuning of CatBoost or ResNet outperformed the choice between tree-based and neural model families.

Categorical predictors introduce a related modelling risk. Prokhorenkova et al. (2018) show that conventional target encoding can allow an observation's outcome to influence its own encoded value, producing target leakage. CatBoost mitigates this by calculating ordered target statistics from preceding observations in a random permutation. Together, these findings support comparing tree-based ensembles with linear and logistic regression baselines.

1.4 Feature Engineering

Recent studies engineer features representing routes, order complexity and operating conditions rather than solely relying only on raw fields. De Araujo and Etemad (2021) supplement origin and destination coordinates with Haversine distance in a model trained on 3.25 million parcel deliveries. Salari et al. (2022) represent order complexity using the number of distinct products and total item quantity, and combine order hour and weekday with recent throughput and outstanding workload. Their forecasts showed that order timing captures operational cut-offs and batching schedules rather than seasonality alone. Preethi et al. (2021) build on this idea directly, modelling pendency, the backlog carried over from recent order and shipment volumes, and find that this backlog remains elevated through and beyond major sales events rather than clearing immediately once the event ends. Gibellini et al. (2024) calculate historical delivery-time summaries across combinations of shipment location, destination and event type, which improved CatBoost accuracy in their setting. K  p et al. (2024) convert delivery event logs into elapsed times between operational checkpoints, decomposing total lead time into processing, handling and shipping durations to identify where delays arise. For a forecast made when an order is placed, however, these durations are outcomes rather than predictors, so only completed stages could inform a later forecast update.

1.5 Evaluation Metrics

For imbalance classifications where late deliveries are rare, accuracy alone can be misleading. Gaudreault and Branco (2024) therefore recommend using multiple complementary metrics for imbalanced classification, favouring the area under the precision-recall curve and the Matthews Correlation Coefficient over F1 or geometric mean when labels are noisy. PR-AUC evaluates performance across possible thresholds, while precision, recall and F1 describe performance at a selected threshold that should reflect the relative costs of missing a late order against unnecessarily intervening in an on-time one. Kandula et al. (2021) found similar AUC values across resampling and cost-sensitive approaches on two e-commerce delivery datasets, selecting weighted XGBoost mainly for its competitive G-mean and computational efficiency. Van den Goorbergh et al. (2022) showed that resampling techniques like SMOTE did not improve AUROC and systematically inflated minority-class probabilities. Any resampling or class weighting should therefore be compared against an unadjusted model using a test set that retains the natural class distribution.

For lead-time regression, Hodson (2022) argues that neither MAE nor RMSE is inherently preferable. MAE is less affected by exceptionally large errors, while RMSE gives these errors greater influence. Reporting both gives distinct operational perspectives: MAE communicates the typical size of a forecasting error in days, while RMSE indicates whether occasional large misses materially weaken performance.

1.6 Conclusion

Fulfillment performance directly links operational efficiency to customer satisfaction, motivating pre-emptive prediction over retrospective observation. Tree-based ensembles are a well-supported starting point for structured e-commerce data of this kind, provided they are compared against simpler linear and logistic regression baselines rather than assumed to win by default, a caution reinforced by McElfresh et al.'s (2023) finding that tuning within a model family often matters more than the choice between families. Feature engineering should draw on route, timing and order-complexity information available at the point an order is placed, while avoiding features that are themselves outcomes of the delivery process. Evaluation should move beyond accuracy for any imbalanced classification target, and should report complementary metrics whose thresholds and costs are chosen deliberately rather than by convention.

2. Exploratory Data Analysis

2.1 Dataset Overview and Data Quality

The Olist dataset covers Brazilian e-commerce activity across orders, products, customers, sellers, payments, deliveries, reviews and geographic locations. The core dataset contains 112,650 line items across 98,666 orders, with each row representing one item within an order. It covers 95,420 unique customers, 3,095 sellers and 71 named English product categories, together with an Unknown category. Total product sales were approximately R\$13.59 million.

The core transactional structure was reliable: identifiers were unique at their expected levels, and all order items matched valid orders, products and sellers. Missingness was handled according to the analysis rather than through blanket imputation. Of the 98,666 orders, 96,470 (97.0%) had complete delivery and estimated timestamps and were used for lead-time analysis. Missing delivery timestamps largely belonged to incomplete orders, so delivery analyses used only delivered orders with the required timestamps. Missing product categories were retained as Unknown and products without physical measurements were omitted only from weight-related analysis. Review text was not required because customer-experience analysis used review scores, and only the most recent review was retained when an order had multiple reviews.

A temporal inconsistency was identified in 1,359 orders where carrier handoff was recorded before order approval. These records were retained for general analysis but excluded from delivery-stage decomposition since they would produce negative stage durations. The geolocation table contained multiple observations for some ZIP prefixes, so coordinates were aggregated into ZIP-prefix centroids before geographic analysis to prevent row multiplication and distorted results.

2.2 Temporal and Commercial Patterns

Marketplace activity grew substantially over time. Orders rose from 22,693 (Jan-Aug 2017) to 53,774 (Jan-Aug 2018), a 137.0% increase, with product sales growing at a similar 137.3%. November 2017 recorded a pronounced peak of 7,451 orders and roughly R\$1.01 million in sales, consistent with Black Friday, though fewer than two complete calendar years limits confidence in a repeatable seasonal pattern (Dashboard Figure 1). Growth came from higher volume, not larger baskets: average order value (R\$137.75) and items per order (1.14) changed little across the dataset (Dashboard Figure 2).

Within that spike, weekly order volume peaked at 2,915 orders in the week of 20 November 2017 (Dashboard Figure 17). Delivery performance did not fail and recovered immediately. Median delivery time rose from a 10 days baseline to 11-13 days and stayed elevated for six consecutive weeks, while cumulative backlog kept building after the spike subsided, peaking at 4,429 orders the following week (Dashboard Figure 18). The relationship between weekly order volume and delivery time is nonetheless weak overall ($r = 0.13$) (Dashboard Figure 19): the single worst week for delivery time, a 15 days median in late January 2018, occurred during a comparatively quiet period of only 1,561 orders, not during the volume peak. This indicates that delay is better explained by accumulated backlog than by concurrent order volume, since a demand spike's effect compounds over subsequent weeks rather than resolving immediately.

2.3 Customer, Seller and Product Distributions

Marketplace activity was geographically and commercially concentrated. São Paulo accounted for 41.9% of orders, while São Paulo, Rio de Janeiro and Minas Gerais

together represented approximately 66.6% (Dashboard Figure 3). The ten largest product categories accounted for 63.6% of items sold, with `bed_bath_table` leading by item volume and `health_beauty` generating the highest product sales (Dashboard Figure 4). Seller activity was even more concentrated, with the top 10% of sellers generating 67.6% of product sales, well above the equal-distribution line (Dashboard Figure 5). This concentration however does not extend to delivery performance, since late delivery rate is essentially flat across seller-volume deciles (8.3% to 8.6%). Repeat purchasing was low at 3.05% of customers (Dashboard Figure 6), and this rate is itself linked to delivery performance. Product weight, order composition and payment method also relate to delivery time, as detailed under Section Candidate 1, where they support the lead-time prediction target.

2.4 Correlation and Relationship Exploration

Customer reviews were generally positive: the average review score was 4.10, with 58.1% five-star and 14.2% low ratings (Dashboard Figure 7). However, low-rating rates spike during periods of elevated delivery delays (Dashboard Figure 8). Delivery timeliness is the feature most correlated with dissatisfaction: 54.1% of late deliveries received a low rating, compared with 9.2% on-time or early orders, making late deliveries roughly 5.9 times more likely to receive negative feedback (Dashboard Figure 9). Whether an order was late ($r = -0.365$) and total delivery days ($r = -0.334$) correlate far more strongly with poor ratings than route distance alone ($r = -0.058$), suggesting the delivery promise itself, not raw distance, is the more informative feature (Dashboard Figure 32). These results show strong association between delivery performance and customer satisfaction, adding impetus to explore further.

A composite freight-to-price ratio, combining distance, weight and carrier pricing into one feature, showed a flat late rate (7.8% - 8.4%) across every bucket tested, offering a useful negative result for Phase 2 feature pruning (Dashboard Figure 22). Olist's promised delivery window is itself not stationary where it correlates more strongly with a trailing eight weeks moving average of actual performance ($r=0.45$) than with live backlog ($r=-0.26$), indicating the estimate reacts to a lagging baseline rather than real-time capacity (Dashboard Figure 23). A detail directly relevant to how Candidate Problem 1's classification target is defined.

2.5 Candidate 1 Problem Exploration

The first candidate is to predict an order's delivery lead time, framed as both a regression on the purchase-to-delivery duration and a classification of on-time versus late against Olist's estimated delivery date, supporting a pre-emptive use case since the outcome can be anticipated before it occurs rather than only observed afterward.

Customer-seller distance shows the strongest relationship of the features tested. Mean delivery time more than triples across distance sextiles, from 5.9 days to 18.5 days (Dashboard Figure 12a), while late-delivery rises more gently, from 6.3% to 11.6% (Dashboard Figure 12b), indicating Olist's estimates already partly adjust for distance. Purchase timing is also informative with Friday orders taking 14.3% longer than Sunday (13.12 versus 11.47 days) (Dashboard Figure 27), an effect localised to the handling stage rather than shipping (Dashboard Figure 21). Payment method separates cleanly as well, with `Boleto` orders averaging 13.04 days against 11.88 for credit card (Dashboard Figure 15a), concentrated in the processing stage and consistent with `Boleto`'s manual bank-confirmation step (Dashboard Figure 16). Order complexity runs counter to intuition with multi-seller orders averaging 8.66 days against 11.77 days for single-seller orders, and multi-item orders are also modestly faster than single-item orders (11.36 versus 12.17 days) (Dashboard Figure 28). Order weight, by comparison, showed only a weak relationship with delivery time (Dashboard Figure 29).

Every one of these features is available when the order is placed, so this target carries no leakage risk from its strongest predictors. The main open question for Phase 2 is defining the classification label itself. As established in Section 2.4, Olist's estimate is not fixed, so a decision is required between classifying against Olist's own shifting estimate or a fixed threshold independent of it. The classification target is also imbalanced (late rates of roughly 6-12% across segments), so accuracy alone would be a misleading evaluation metric. Precision, recall and F1 on the late class or a threshold-adjusted probability output will be considered. The prediction is intended for Olist's logistics and fulfilment operations team, who could use a flagged at-risk order to expedite a shipment or proactively notify the customer before the delivery promise is breached. Overall, the dataset offers substantially more pre-delivery predictive signal for lead time than for low-rating prediction directly, which is why delivery lead time prediction will be prioritised as the primary Phase 2 problem.

2.6 Candidate 2 Problem Exploration

The second candidate is to classify orders at risk of receiving a low customer rating, allowing Olist to prioritise customer-experience interventions. A low rating is a latest review score of one or two, accounting for 14.2% of reviewed orders (Dashboard Figure 7).

Numerical features were assessed using Spearman correlation, while categorical features were assessed using Cramér's V and group-level rates. Days late had the strongest numerical relationship with low ratings ($\rho = 0.374$). Orders delivered more than seven days late were 8.7 times as likely to receive a low rating as orders more than seven days early (Dashboard Figure 30), a starker escalation than the 5.9 times seen for any lateness mentioned in Section 2.4, since this compares only the most extreme bands. Multi-item orders were 2.2 times as likely to receive a low rating as single-item orders, while multi-seller orders were 3.4 times as likely as single-seller orders (Dashboard Figure 31).

However, actual days late is known only after delivery and cannot support pre-emptive intervention; using it pre-delivery would introduce target leakage. The strongest early-available indicators also have limited coverage: multi-item orders accounted for only 19.3% of low-rated orders, and multi-seller orders were even less common. Other plausible drivers, including product quality, incorrect items, packaging and customer service, are not available as structured features. The candidate has clear business value, but the dataset provides limited evidence that low ratings can be predicted effectively before delivery. Further modelling would be required to assess whether combining the available early features produces useful out-of-sample performance.

2.7 Key Insights and Patterns Discovered

Overall, the EDA depicts an expanding but concentrated marketplace. Growth came from higher order volume rather than larger baskets, activity was concentrated across states, product categories and sellers, and most customers made only one observed purchase. Reviews were generally positive, but late delivery had by far the clearest relationship with dissatisfaction, and this effect is not confined to the order itself: backlog peaks after demand spikes rather than during them and stays chronically elevated even outside holiday periods, while total lead time is dominated by shipping transit (~74% on average) (Dashboard Figure 10), the only stage whose duration correlates with order volume, pointing to carrier capacity rather than internal processing as the bottleneck under load. Together with the purchase-timing and promise-window findings above, this identifies delivery performance as the clearest standalone analytical direction for Phase

2, while low-rating classification overlaps substantially with delivery and is constrained by missing non-delivery predictors such as product quality and customer service.

3. Links

Github Link:

https://github.com/YuanJiayi/Team6_IT5006_Ecommerce_Analytics_AY2627Sem1

Dashboard Link: <https://team6-it5006-ecommerce-analytics-ay2627sem1.streamlit.app/>

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